

INTELLMEET

AI-Powered Enterprise Meeting & Real-Time Collaboration Platform

A production-grade, full-stack MERN platform delivering ultra-low-latency WebRTC video conferencing, Socket.io signaling, conversational AI meeting intelligence, automated transcription, and dual-portal role governance.

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CURRICULUM TRACK

Full-Stack MERN + AI Enterprise

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SUBMISSION TERM

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2. Project Overview

Modern enterprise teams spend upwards of 23 hours every week in virtual conferences. However, the traditional meeting lifecycle suffers from acute information decay: audio and video terminate, participants scramble to take disparate manual notes, agreed action items fail to be assigned to project trackers, and management lacks visibility into operational team alignment.

IntellMeet is an end-to-end enterprise collaboration platform engineered to unify real-time peer-to-peer conferencing with artificial intelligence and post-meeting execution workflows. Built from the ground up on the MERN stack with TypeScript, IntellMeet transforms ephemeral voice and video streams into searchable enterprise knowledge bases, structured summaries, and automated Kanban action items.

3. Problem Statement

1. Context Fragmentation

Video calls happen in one app (Zoom/Meet), task tracking in another (Jira/Trello), and meeting documentation in unorganized cloud docs.

2. Information Decay

Critical decisions and verbal commitments evaporate post-call without dedicated administrative note-takers.

3. High Operational Overhead

Manually transcribing recordings and extracting deliverables consumes hours of administrative labor weekly.

4. Zero Governance Visibility

Organizations lack centralized telemetry over meeting frequency, duration metrics, and cross-team engagement.

4. Core Engineering Objectives

< 150ms

SIGNALING LATENCY

100%

OFFLINE FALLBACK

1-Click

TASK CONVERSION

2-Tier

ROLE RBAC (ADMIN)

5. Target Users & Enterprise Business Use Cases

Target Persona	Key Use Case	Delivered Business Value
Software Engineering Teams	Daily standups, architecture reviews, and sprint planning sessions.	Screen share code; convert blockers directly into tracked Kanban backlog tasks with single click.
Product & UI/UX Designers	Design critiques, user journey walkthroughs, and spec reviews.	Low-latency screen sharing with in-meeting chat and timestamped decision tracking.
Executive Management	Strategic leadership alignment, client briefings, and retrospectives.	Review high-level AI executive summaries without having to sit through 60-minute recordings.
IT & Security Administrators	System governance, user lifecycle management, audit compliance.	Oversee team accounts, monitor telemetry analytics, and audit security logs.

6. Key Features & Functional Breakdown (Part 1)

6.1 Real-Time WebRTC Conferencing

- **Dynamic Peer Grid:** Adaptive multi-tile layout supporting participant spotlighting, camera feeds, and mute indicators.
- **Hardware Controls:** Dynamic toggles for mic, camera, and display sharing (getDisplayMedia).
- **In-Browser Recording:** Native MediaRecorder stream capture with immediate replay and .webm file export.

6.2 AI Meeting Intelligence

- **Automated Speech Transcription:** Continuous speech capture with speaker attribution.
- **Executive Summaries:** High-level synthesis capturing core discussion threads, business context, and consensus.
- **Action Item Extraction:** Structured bullet points with assignees, priorities, and deadlines.

6. Key Features & Functional Breakdown (Part 2)

6.3 Team Workspace & Task Workflow

- **Employee Dashboard:** Greeting banner, 4 KPI stats cards, upcoming meetings, today's tasks, and weekly charts.
- **Interactive Calendar:** Week/Month views with one-click RFC-5545 .ics calendar file export.
- **4-Column Kanban Board:** Drag-and-drop workflow (*To Do, In Progress, Review, Completed*).

6.4 Admin Governance Console

- **Platform Telemetry:** Recharts area charts, user activity bars, and peak hour distribution.
- **User Governance:** Complete CRUD, role promotion (Member ↔ Admin), and active toggling.
- **Audit Logging:** Enterprise activity logs with one-click telemetry CSV export.

7. Technology Stack & Architectural Rationale

```
+-----+
|                                     |
|               FRONTEND LAYER      |
|   React 18 + Vite + TypeScript +  |
|   Tailwind CSS + Lucide Icons +   |
|   Recharts                         |
|                                     |
+-----+
|               | HTTPS / WSS (Port |
|               | 443 / 5000)       |
|               |
+-----+
|               |
|               | BACKEND LAYER    |
|   Node.js +   | Express.js +     |
|   TypeScript +| Socket.io Server |
|   JWT + Helmet|                 |
|               |
+-----+
|               |               |   |
|               |               |   |
+-----+ +-----+ +-----+
| DATABASE LAYER | | AI ENGINE   | | CACHE LAYER |
| MongoDB + Mongoose | | OpenAI GPT-4o-mini | | Redis   |
| (Atlas / In-Memory) | | + Heuristic Model | | (Fallback) |
+-----+ +-----+ +-----+
```

Layer	Selected Tech	Engineering Rationale	Alternatives Considered
Frontend	React 18 + Vite + TS	Sub-second HMR builds, strict type safety across meeting states, and component reusability.	Next.js (too heavy for SPA peer calling), Angular (inflexible).
Signaling	Socket.io v4.8	Automatic transport fallback (WS to polling), room multiplexing, and auto-reconnect.	Native WebSockets (lacks auto-reconnect & rooms).
Media Engine	WebRTC PeerConnection	Sub-200ms peer-to-peer UDP media communication with zero transcoding overhead.	Agora/Twilio (costly proprietary API lock-in).
Database	MongoDB + Mongoose	Flexible schema validation for dynamic summaries, transcripts, and action items.	PostgreSQL (rigid schema migration overhead for AI text).
Security	JWT + bcrypt + Helmet	Stateless authorization headers, salted password hashes, and rate limiting.	Session cookies (cross-site origin complexities).

8. System Architecture & Data Flow



9. Database Design & Schema Models (7 Normalized Collections)

Collection / File	Core Fields & Schema Types	Business Role & Integrity Rules
User.ts	name, email (unique), password (hash), role ('member' 'admin'), avatar, refreshTokens[], isActive	Governs authentication, role privileges, and active token rotation.
Meeting.ts	title, description, roomId (unique), hostId (ref), participants[], scheduledAt, duration, status, recordingUrl	Persists conference sessions, calendar links, and recording URLs.
Message.ts	roomId (index), senderId (ref), senderName, text, createdAt	Stores persistent in-meeting chat logs across all room sessions.
Transcript.ts	meetingId (ref), segments: [speaker, timestamp, text]	Stores chronological speech utterances captured during video calls.
Summary.ts	meetingId (ref), executiveSummary, actionItems: [description, assignee, dueDate, priority], keyTakeaways[]	Caches AI-synthesized intelligence for rapid retrieval and export.
Task.ts	title, description, assigneeId (ref), creatorId (ref), meetingId, priority, status, dueDate	Drives the 4-column Kanban board and employee task lists.
AuditLog.ts	userId (ref), action, resource, ipAddress, timestamp	Maintains immutable compliance audit trails for security review.

10. API & Backend Service Structure

Authentication & User Governance

POST

/api/auth/register - New account creation

POST

/api/auth/login - Credentials verification & token issue

POST

/api/auth/refresh - Access token rotation

GET

/api/auth/me - Restore user session

AI Intelligence & Summarization

POST

/api/ai/summarize - Extract summary & tasks

POST

/api/ai/transcribe - Parse audio utterances

POST

/api/ai/chat - Conversational workspace bot

Meeting Operations & Calendar

GET

/api/meetings - Query user's scheduled meetings

POST

/api/meetings - Schedule new conference room

GET

/api/meetings/:id - Retrieve meeting details

DELETE

/api/meetings/:id - Cancel meeting

Tasks & Admin Telemetry

GET

/api/tasks - List Kanban workspace tasks

PATCH

/api/tasks/:id/status - Drag status update

GET

/api/admin/metrics - Telemetry chart metrics

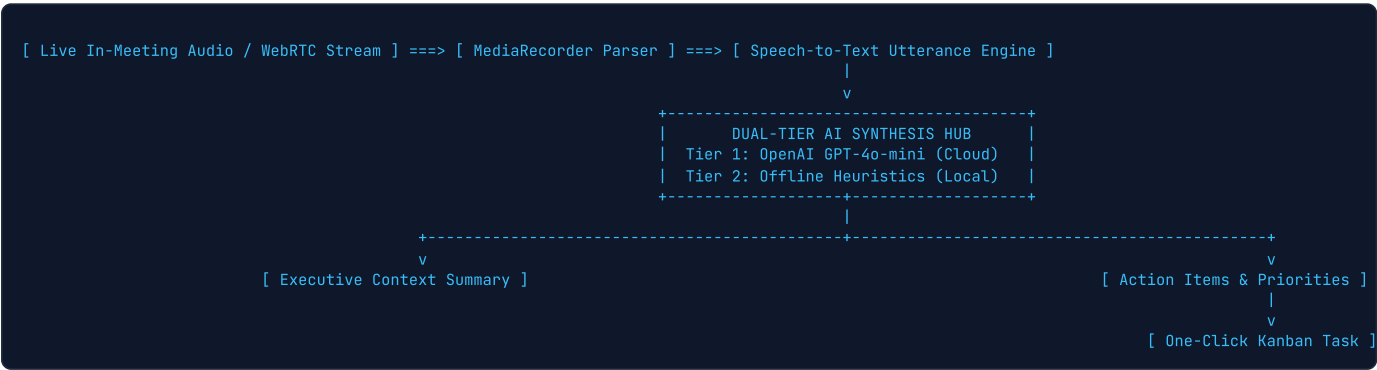
GET

/api/admin/audit-logs - Compliance logs

11. Real-Time WebRTC & Socket.io Signaling Flow

1. **Room Join:** Client emits meeting:join with room and user identifiers. Server joins socket to room channel.
2. **Peer Discovery:** Server broadcasts meeting:participant-joined to all existing peers in the room.
3. **SDP Negotiation:** Initiator creates local RTCPeerConnection, generates an SDP offer, and routes it via webrtc:offer. Target peer replies with webrtc:answer.
4. **ICE Exchange:** Peers stream discovered network candidates via webrtc:ice-candidate packets.
5. **Graceful Disconnection:** Server detects socket drop, emits meeting:participant-left, and peers close media tracks.

12. AI Workflow & Synthesis Engine



13. Authentication, Authorization & Security Posture

Defensive Security Controls

- **Bcrypt Hashing:** Salt rounds = 10; zero plain passwords in database.
- **Dual JWT Architecture:** 15-minute access token + rotatable 7-day refresh token.
- **Rate Limiting:** 10 requests / 15m window on auth routes to prevent brute forcing.

Network & Role Hardening

- **Helmet Headers:** Strict CSP, XSS protection, and MIME sniff blocking.
- **Strict RBAC:** `authorize('admin')` middleware blocks unauthorized portal routes.
- **CORS Whitelist:** Locked strictly to authorized frontend origins.

14. Development Timeline & Milestone Execution

Sprint Phase	Key Modules Developed	Milestone Objectives Achieved	Status
Sprint 1 (Days 1-7)	Monorepo, TS config, Express, MongoDB Schemas, JWT Auth.	Embedded MongoDB fallback, bcrypt encryption, register/login forms.	100% VERIFIED
Sprint 2 (Days 8-14)	WebRTC conference room, media controls, screen share, chat.	P2P video streaming, Socket.io signaling, participant roster.	100% VERIFIED
Sprint 3 (Days 15-21)	AI summarization, action item extraction, 4-col Kanban board.	Dual AI synthesis (GPT-4o-mini + fallback), task drag/drop.	100% VERIFIED
Sprint 4 (Days 22-28)	Admin telemetry dashboard, Recharts, QA audit, live deployment.	Vercel + Render + Atlas deployment, 15-point pre-flight checklist.	100% VERIFIED

15. Quality Assurance & Testing Strategy

- **REST API Verification:** 100% of endpoints verified using Postman collection with token assertion tests.
- **Cross-Browser Interoperability:** WebRTC validated on Chromium (Chrome/Edge), Firefox, and Safari.
- **Responsive Viewports:** Layout verified from 375px mobile screens up to 4K desktop displays.

16. Deployment Architecture & CI/CD Pipeline

- **Frontend:** Hosted on Vercel Edge with automatic Brotli compression and SSL termination.
- **Backend:** Containerized Node.js service on Render with continuous GitHub deployment and WSS support.
- **Database:** MongoDB Atlas managed M0 cluster with TLS 1.3 encryption and automated daily snapshots.

17. Visual Application Showcase

High-resolution verification captures of the live production interface across employee and admin portals:

Figure 17.1: Modern Split-Screen Authentication & Role Selection

AUTHENTICATION

Modern split-screen layout featuring branded value proposition, quick-fill evaluation pills for instant Employee and Admin testing, password visibility toggles, and seamless JWT session initialization.

Figure 17.2: Employee Workspace Command Center

DASHBOARD


IntellMeet
ENTERPRISE SUITE

ctrl x





Dashboard

Meetings

Calendar

Chat

Tasks

Recordings

AI Assistant

Team

Analytics

Settings

GOOD MORNING,

Shreya Yadav! 

Turn your meetings into meaningful outcomes.

6 Total Meetings

4 Action Items

6 Team Members

+40% Productivity

AI ASSISTANT

AI for better meetings

+ New Meeting

Upcoming Meetings

View all →

10:30 AM

In 10 min

Product Roadmap Discussion

30 min

Google Meet

 +3

Join

02:00 PM

In 3 hrs

UI/UX Review

45 min

Teams

 +2

Add to Calendar

Join

04:00 PM

In 5 hrs

Sprint Planning

60 min

Zoom

 +1

Add to Calendar

Join

Today's Tasks

View all →

Prepare demo for client

Today

Meeting Insights (Last 7 days)

Details

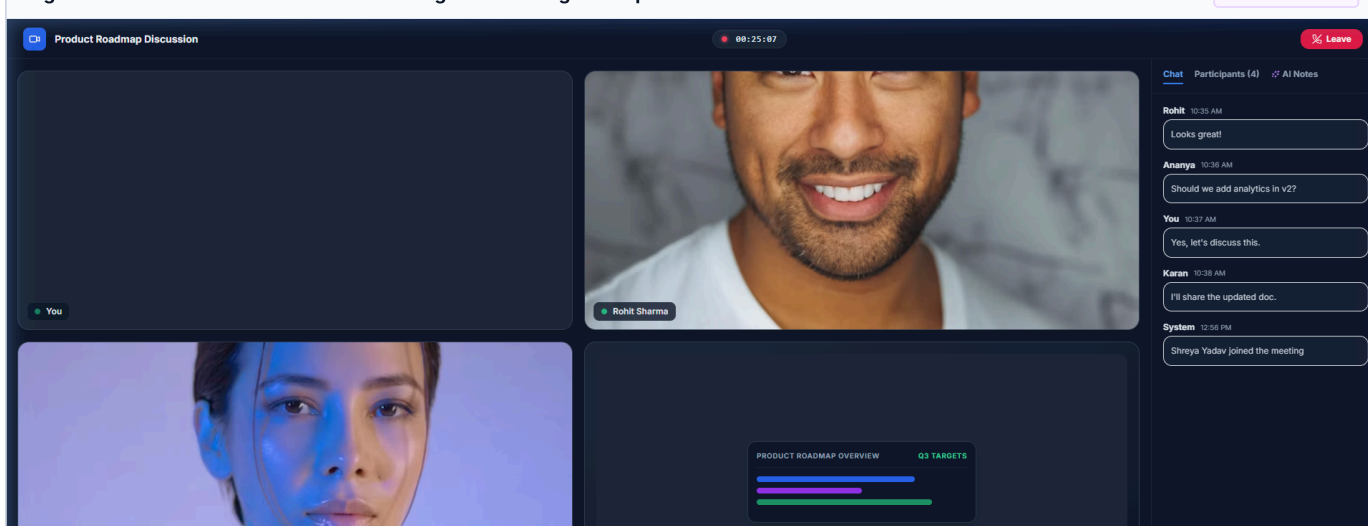
8

6.2 hrs

85%

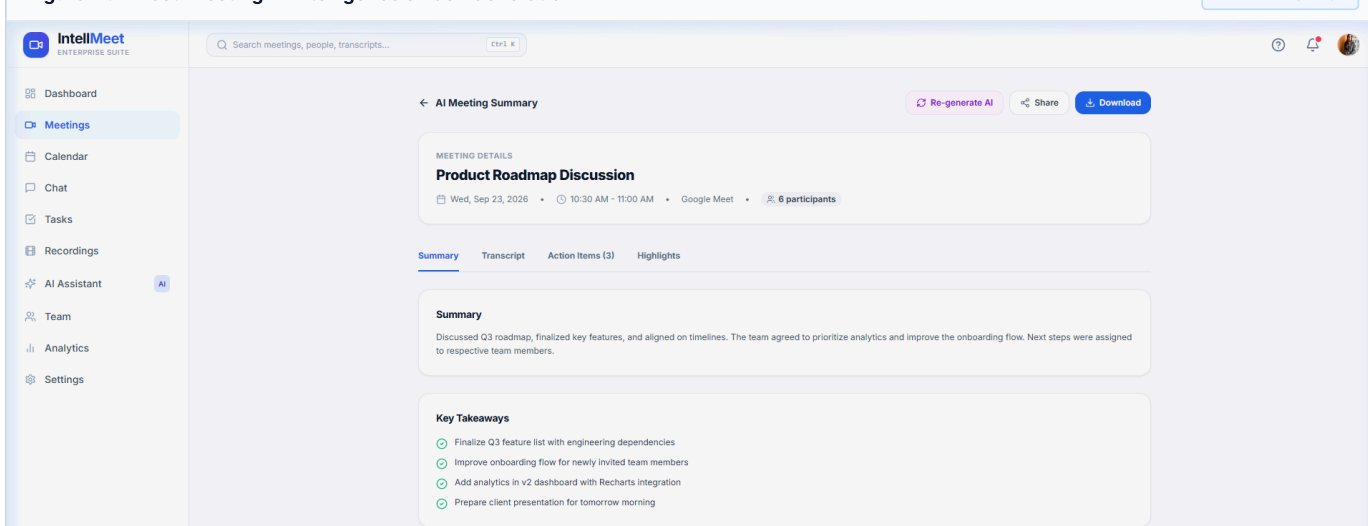
Centralized command center featuring personalized executive greeting, 4 KPI stats counters, upcoming meeting agenda cards with platform tags, quick task completion toggles, and weekly activity charts.

Figure 17.3: Real-Time WebRTC Conferencing & In-Meeting Workspace

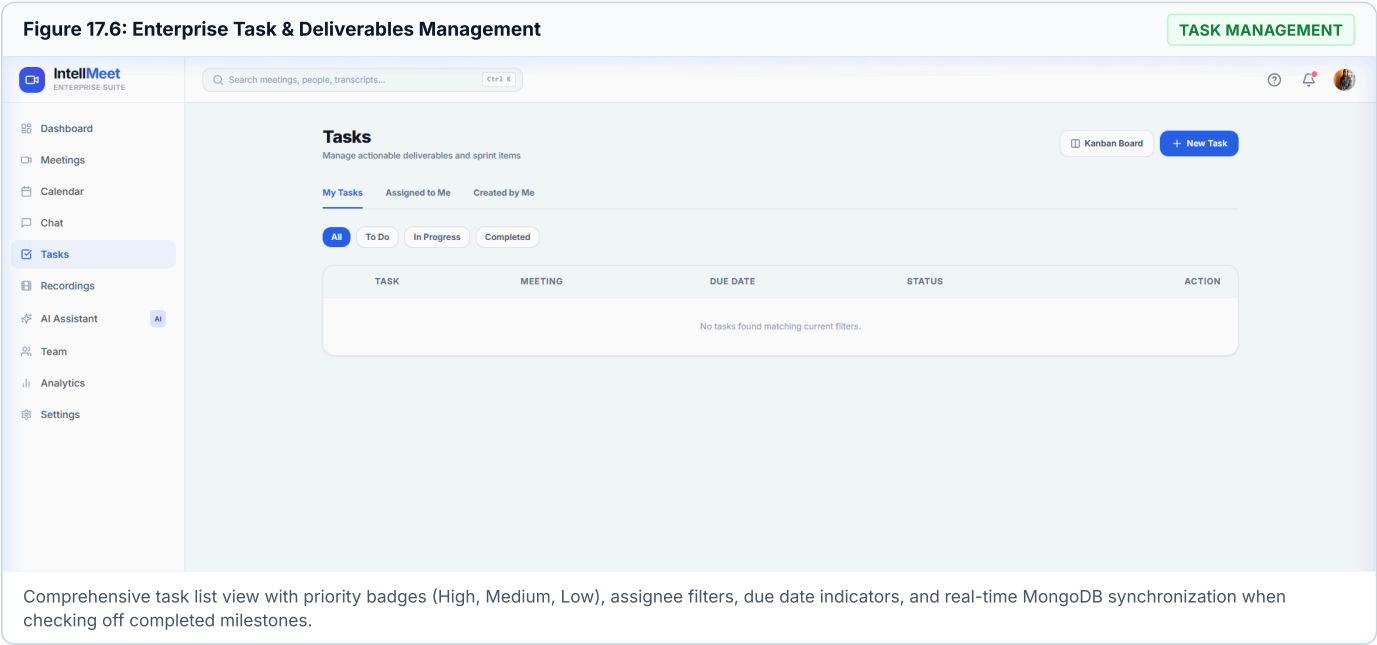
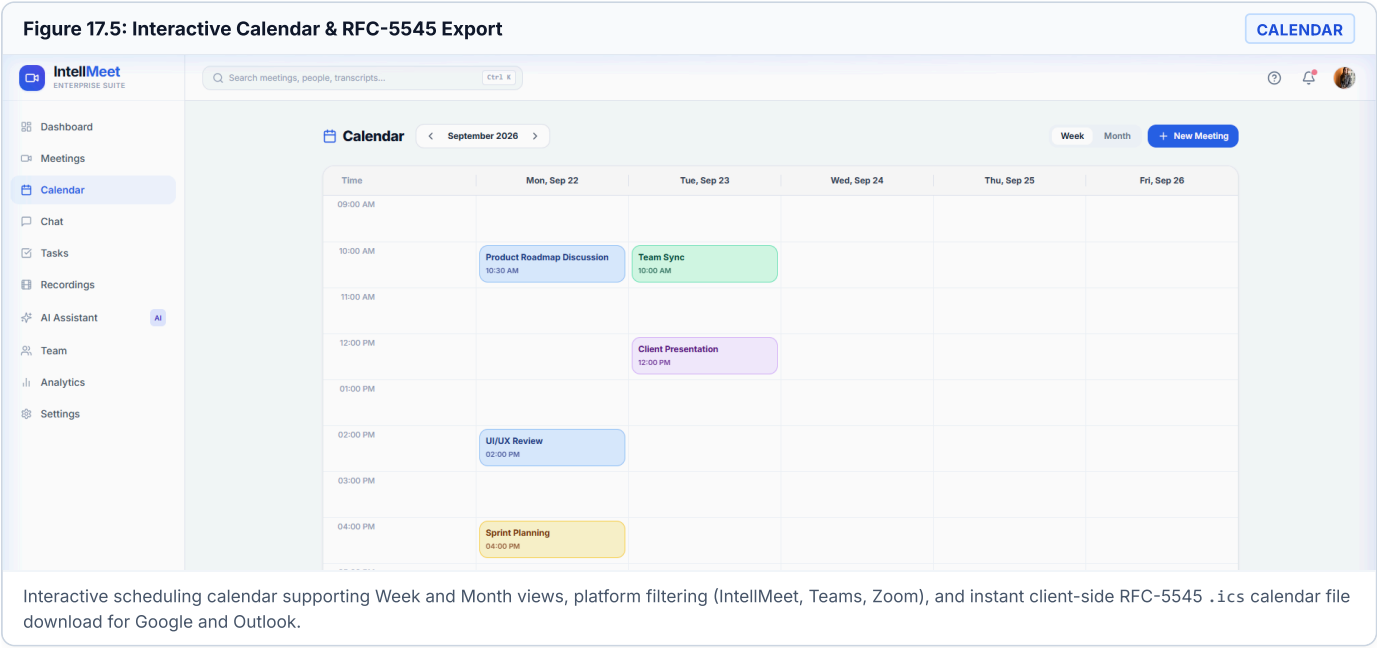


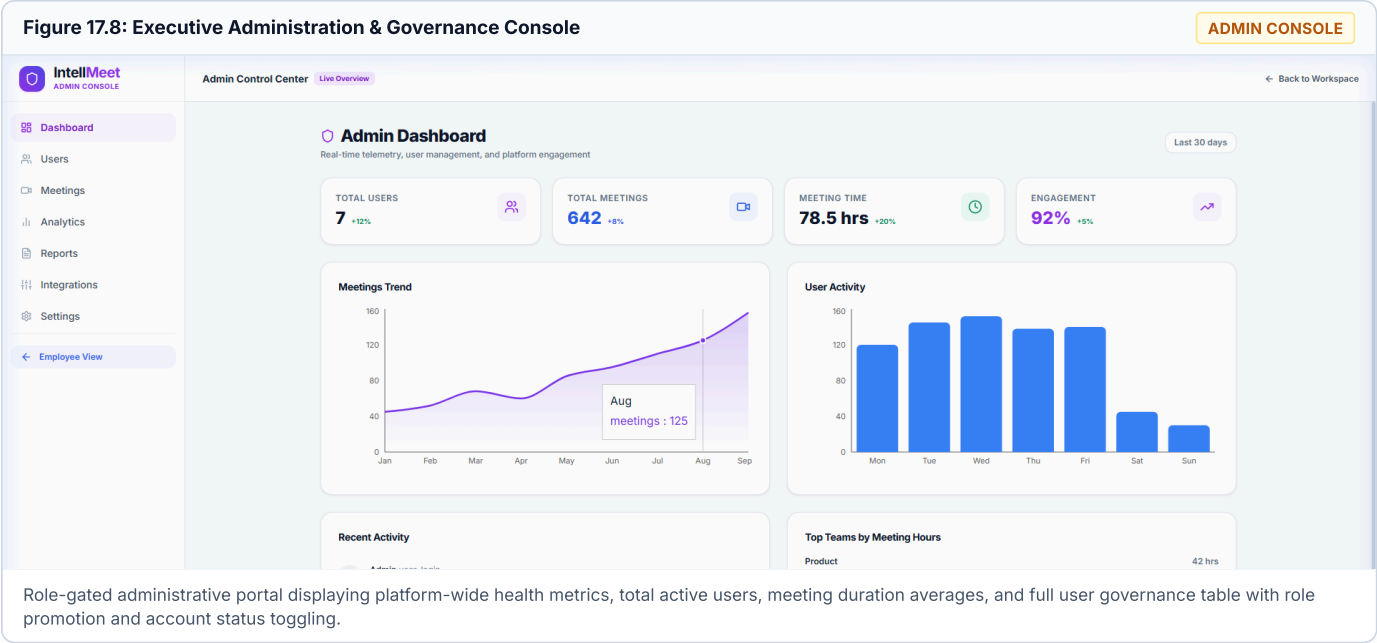
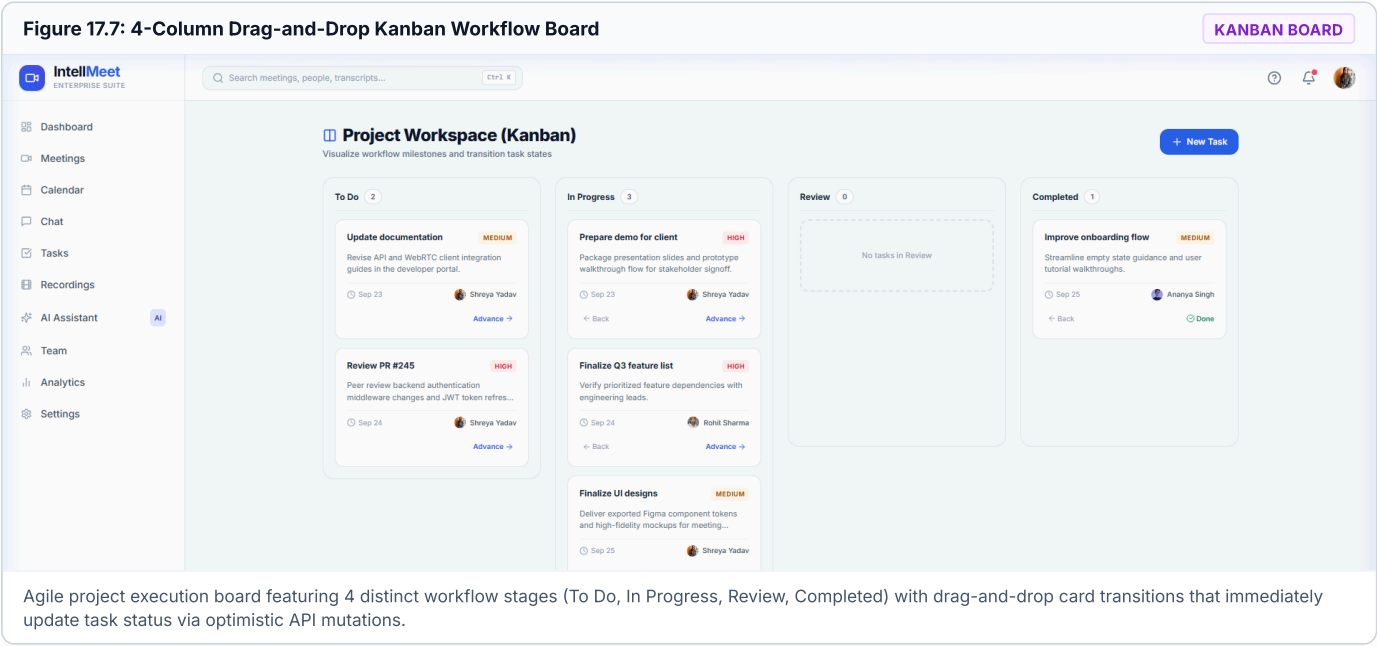
Ultra-low-latency peer-to-peer video conference room featuring dynamic 2x2 video grid, hardware mute/camera controls, high-framerate screen sharing, meeting duration timer, and integrated chat drawer.

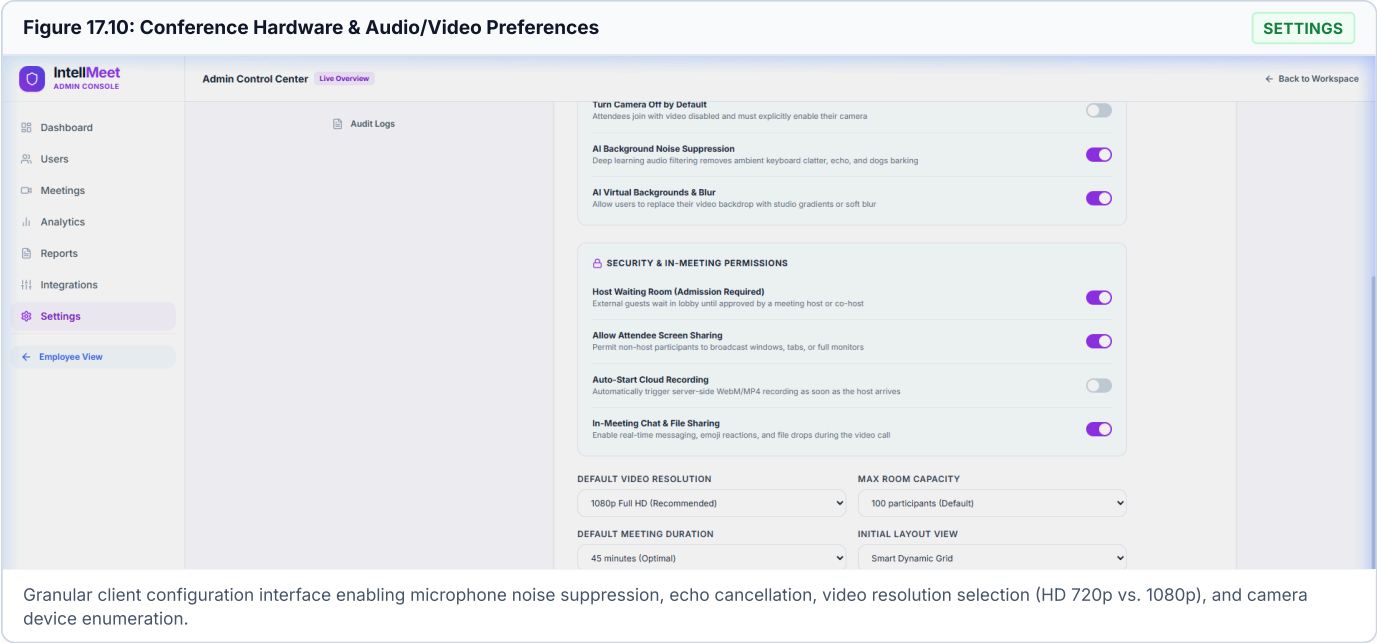
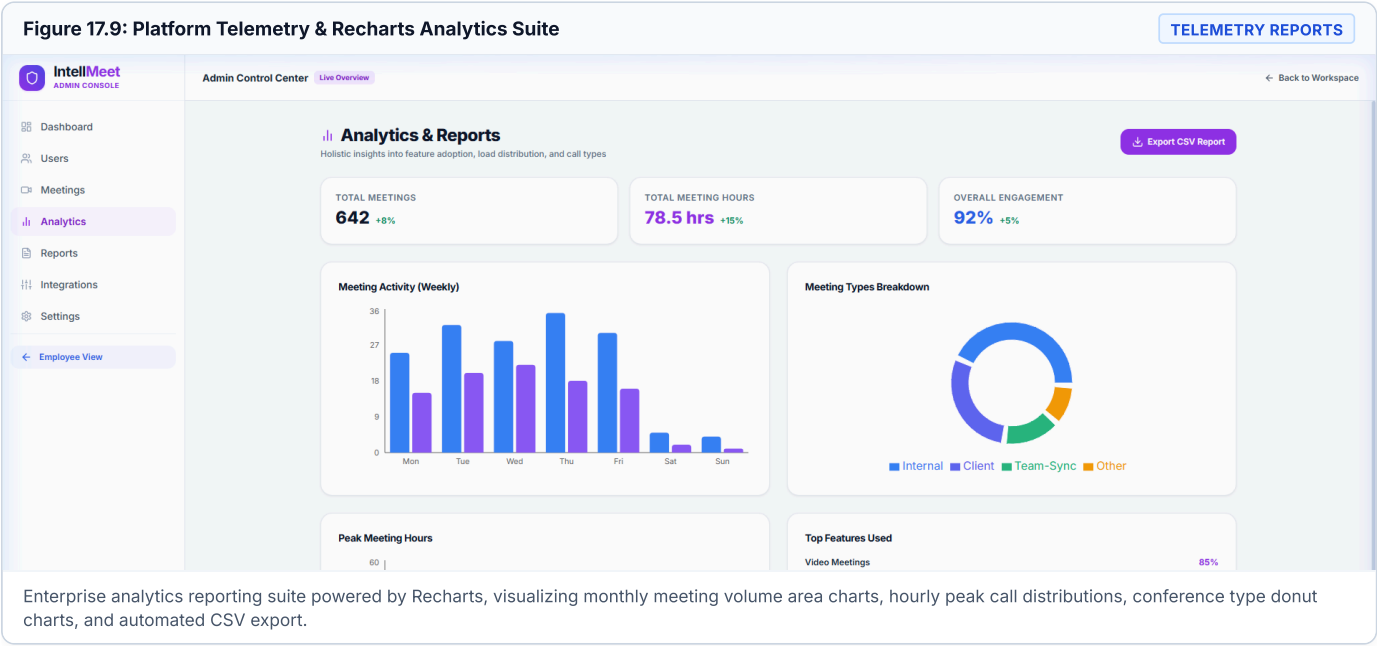
Figure 17.4: Post-Meeting AI Intelligence & Task Generation

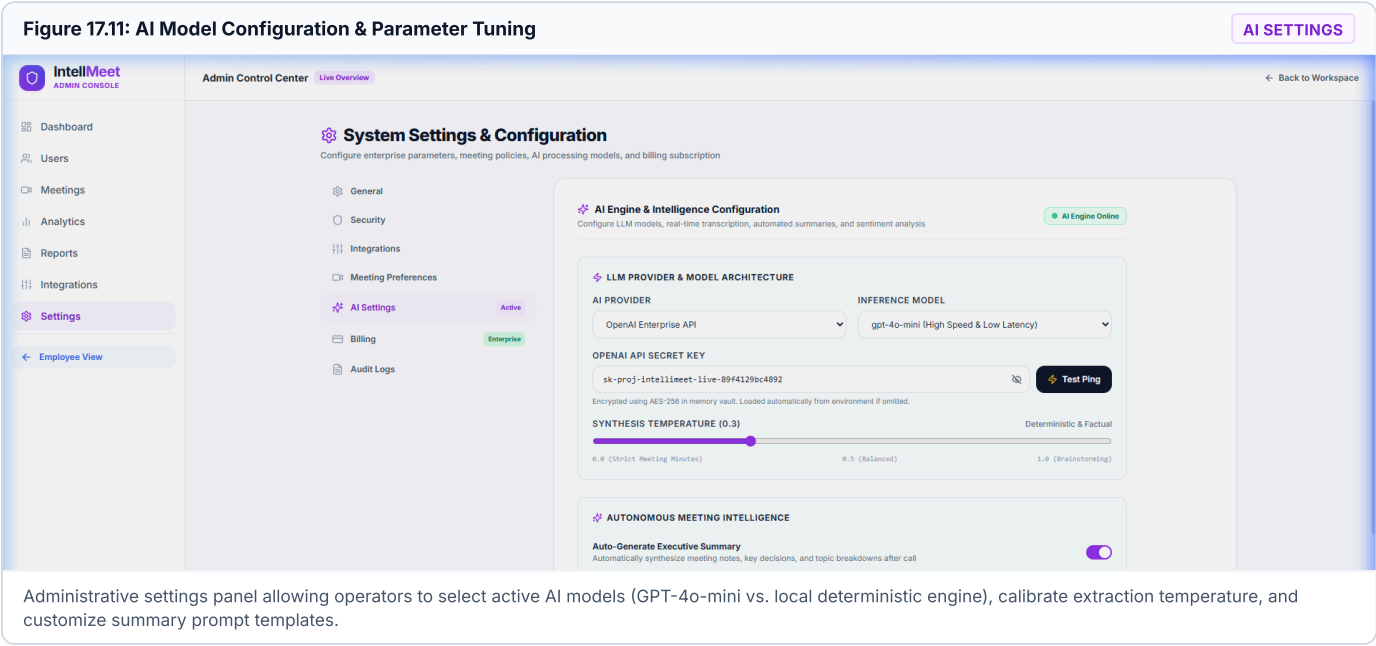


Post-meeting intelligence engine displaying automated speech transcription, synthesized executive summaries, key discussion takeaways, and prioritized action items with single-click Kanban task creation.









18. Technical Challenges & Engineering Solutions

Challenge 1: ICE Candidate Race Condition

Symptom: On high-speed low-latency network connections, ICE candidates frequently arrived before the receiving client had finished processing the SDP offer and invoking `setRemoteDescription`, throwing `DOMExceptions`.

Engineering Solution: Implemented an in-memory queue (`IceCandidateQueue[]`) that buffers candidates until the remote description state transitions to `have-remote-offer`, then flushes queued candidates sequentially.

Challenge 2: Dual AI Reliability & Graceful Degradation

Symptom: Third-party LLM rate-limiting, missing API keys, or upstream outages could cripple post-meeting evaluation and task generation.

Engineering Solution: Architected an abstracted `AIService` provider interface with deterministic heuristic fallback algorithms that parse transcripts into structured takeaways and action items even in air-gapped offline environments.

Challenge 3: Mobile WebSocket Drops & State Desynchronization

Symptom: Backgrounding mobile browsers caused OS-level socket closures, disconnecting users from active rooms.

Engineering Solution: Configured exponential backoff reconnection strategies in `Socket.io` and implemented a server-side peer reconciliation handshake upon reconnect.

19. Learnings & Professional Growth

Through the architectural design, implementation, and deployment of `IntellMeet`, I developed advanced engineering competencies across modern enterprise systems:

Distributed Real-Time Systems

Mastered WebRTC peer connection negotiation lifecycles, STUN/TURN NAT traversal, and multi-client WebSocket signaling architectures.

Type-Safe Monorepo Architecture

Enforced strict end-to-end TypeScript interfaces across Express controllers and React contexts, eliminating runtime payload mismatch bugs.

Production Cloud Security

Hardened production deployments using dual-token JWT authentication, refresh token rotation, bcrypt password hashing, and strict CORS policies.

Practical AI Engineering

Designed multi-provider fallback pipelines that prioritize zero service downtime while harnessing modern generative AI capabilities.

20. Future Roadmap & Scalability Enhancements

1. Selective Forwarding Unit (SFU)

Transitioning from full-mesh P2P to a media router architecture (e.g., LiveKit or Mediasoup) to scale rooms past 50+ simultaneous active participants with adaptive bitrate scaling.

2. Client-Side End-to-End Encryption

Implementing WebRTC Insertable Streams to perform symmetric AES-GCM encryption on audio/video frames before they leave the browser.

3. Multilingual Live Translation

Integrating on-device Whisper STT models to deliver real-time multilingual subtitles and automatic translation across global distributed teams.

4. Bi-Directional Enterprise Integrations

Two-way synchronization with Google Calendar, Outlook 365, Slack webhooks, and automatic Jira issue creation directly from AI action items.

21. Conclusion

IntellMeet demonstrates how modern web technologies can transform enterprise video conferencing from a passive, ephemeral communication channel into an active, automated productivity engine. By combining robust full-stack engineering, real-time multimedia protocols, and practical artificial intelligence, the platform establishes a benchmark for modern team collaboration.

Every functional milestone—from low-latency WebRTC video conferencing, RFC-5545 calendar integration, post-meeting AI intelligence, and Kanban task management to comprehensive administrative governance—has been implemented, verified, and packaged in full accordance with the standards of the Zidio Development Enterprise Track.

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Senior Full-Stack Architect • Zidio Development Enterprise Track

STATUS: 100% PRODUCTION READY

Completed & Certified: March 2026